States Of Matter

I. Introduction

The kinetic theory describes matter as a large number of small particles (atoms or molecules), all of which are in constant, random motion. The rapidly moving particles constantly collide with each other and/or with the walls of the container. Kinetic theory explains properties such as pressure, temperature, viscosity, thermal conductivity, and volume.

Here are the 4 main ideas of kinetic theory (also known as 'the particle model')
• Matter is made up of particles which are in constant random motion (vibrate, rotate, translate)
• The particles are separated by great distances
• The particles collide perfectly elastically
• The temperature of a substance is related to the molecular velocity (speed).

In this activity, you will demonstrate the properties of various types of matter. The demonstration will allow you to change forms of matter, amounts of matter, temperature, and pressure.

II. Procedure

1. Go to the following website : https://phet.colorado.edu/en/simulation/states-of-matter .

2. Click on "Run Now!". Then open the JNLP file.

3. Click on the tab that is labeled "Solid, Liquid, Gas".

4. Using Neon, choose a solid state of matter, record the temperature, and describe the nature of the solid neon in the container.

   Temperature : _____ K, _____C

   Movement of Molecules : _____

   _____

5. Using solid Neon, increase the temperature, record the temperature, and describe the nature of the solid neon.

   Temperature : _____ K _____C

   Movement of Molecules : _____

   _____

6. Using Argon, choose a liquid state of matter, record the temperature, and describe the nature of the liquid argon in the container.

   Temperature : _____ K _____C

   Movement of Molecules : _____

_____

7. Using liquid Argon, increase the temperature, record the temperature, and describe the nature of the liquid argon.

   Temperature : _____ K _____C

   Movement of Molecules : _____

   _____

8. Using Water, choose a gaseous state of matter, record the temperature, and describe the nature of the gaseous water in the container.

   Temperature : _____ K _____C

   Movement of Molecules : _____

   _____

9. Using gaseous Water, increase the temperature, record the temperature, and describe the nature of the gaseous water.

   Temperature : _____ K _____C

   Movement of Molecules : _____

   _____

   _____

10. Choose one substance and draw what the molecules look like in each state

	Solid	Liquid	Gas
Diagram of molecules			
Sentence explaining how molecules will be moving.			

11. Describe how solid water is different to the solids of the other substances.

_____

_____

_____

12. Click on the tab that is labeled "Phase Changes".

13. Using Oxygen, heat up the container and record the stabilizing temperature and pressure without changing the container size.

Heat

Temperature : _____ K _____C Pressure : _____ ATM

Click on Reset.

14. Using Oxygen, cool down the container and record the stabilizing temperature and pressure without changing the container size.

Cool

Temperature : _____ K _____C Pressure : _____ ATM

Design Your Own Experiment

Choose any type of atom or molecule (Neon, Argon, Oxygen, Water).

Create an experiment on your own using the PhET simulation. Describe the experiment below by changing the number of particles (pump), temperature (bucket), or volume (finger).

Determine a conclusion regarding your experiment. Record the following below.

Hypothesis : _____

_____

Independent variable: _____

Dependent Variable: _____

Control variables: _____

Type Of Atom or Molecule : _____

Final Temperature : _____ K

Final Pressure : _____ ATM

Experiment Conclusion : _____

_____